

I Akaike Information Criterion (AIC)

$$-2 \ell(\hat{\beta}, \hat{\sigma}^2) = n \ln SSE_p - n \ln n + n \ln(2\pi) + n$$

Idea: the log-likelihood is biased/optimistic because the estimator are based on our sample, so do a bias correction.

$$AIC = -2 \ell(\hat{\theta}) + 2(p+1)$$

Lower is better

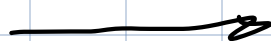
this is the bias

+1 is because we estimate σ^2

Aside: Recap

$$R^2 < R_a^2 < C_p, AIC < BIC$$

biggest model



smallest model

II Bayesian Information Criterion (BIC)

$$BIC = -2 \ell(\hat{\theta}) + \underbrace{(p+1)}_1 \ln n$$

Lower is better

Cramps move quickly
than AIC, so penalizes
more heavily.

AIC

Find model predicts best.

Chooses larger models.

Keep junk predictors.

BIC

Find the true model.

Smaller models.

Drops weak ("good") predictors

How to use:

- Must use same data when comparing models.
- Raw value doesn't matter, use for ranking.
- AIC + BIC might disagree

III

Subset Search Methods

Best subsets: check all 2^{p-1} models

$p-1$

10

20

40

models

1024

$\approx 10^6$

$\approx 10^{12}$

Stepwise Methods

Forward : add the predictor that drops SSE most,
Keep going until some threshold.

Backward : fit full model, drop weakest predictor,
Keep going.

Why we don't use these.

1. Computationally expensive
2. Greedy : always picking best predictor, could miss other pairs that are good together.
3. Invalid p-values - using same data multiple times
4. Unstable
5. Overfitting - artificially good metric

IV Prediction Sum of Squares (PRESS)

1. Let $i=1$
2. Drop obs i from data
3. Fit model on remaining obs.
4. Predict $\hat{y}_i$

5. Take $(y_i - \hat{y}_i)^2$
6. $i = i+1$ and repeat steps 2-5
7. Sum the squared errors

Note: Leave-one-out cross-validation

$$PRESS = \sum_{i=1}^n (y_i - \hat{y}_{i(i)})^2 = \sum_{i=1}^n \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

↑
 $\hat{y}$ for model
fitted w/out
obs i
↑
higher leverage
= harder to predict
when left out of
model

V Warnings

1. Selection bias - always choosing "best" model
2. Multiple testing - selecting for small p-values
3. Multicollinearity - if x_1 and x_2 are highly correlated, model selection chooses one arbitrarily.

VI Solutions

1. Use cross-validation or hold out set
2. Remove correlated predictors first

3. Keep predictors if it makes sense
4. If you have X_1^2 or $X_1 X_2$, keep main effects
5. If $n \gg p$ keep all predictors that make sense

VII Logistic Regression

$$Y_i \in \{0, 1\}$$

$$Y = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 1 \end{pmatrix}_{n \times 1}, \quad X = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & X_1 & X_2 & \dots & X_{p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_1 & X_2 & \dots & X_{p-1} \end{pmatrix}_{n \times p}$$

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

pmf:	value	probability
	1	π_i
	0	$1 - \pi_i$

$$E(Y_i) = \pi_i$$

$$\text{Var}(Y_i) = \pi_i (1 - \pi_i)$$

Why not use LR?

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_{p-1} X_{(p-1)i} + \varepsilon_i \\ &= X_i^T \beta + \varepsilon_i \end{aligned}$$

$$\pi_i = X_i^T \beta$$

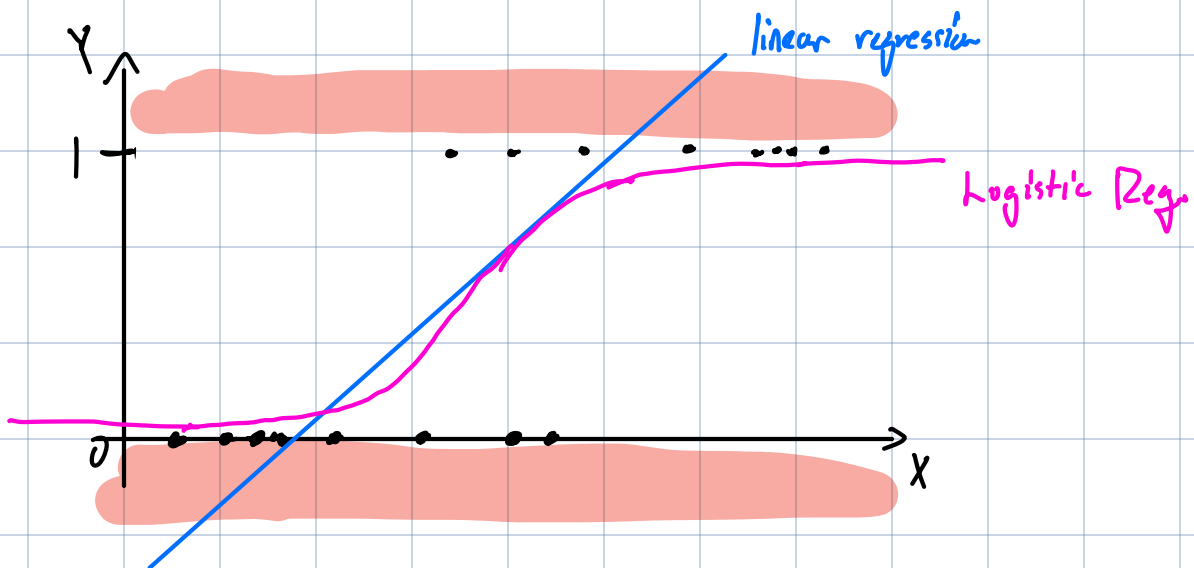
① Errors are binary, not Normal
 $\varepsilon_i = Y_i - X_i^T \beta$

$$\begin{array}{ll} Y_i = 1 & \varepsilon_i = 1 - X_i^T \beta \\ Y_i = 0 & \varepsilon_i = -X_i^T \beta \end{array} \quad \rightarrow \text{binary}$$

② Variance is not constant

$$\text{Var}(Y_i) = \pi_i (1 - \pi_i)$$

③ $0 \leq \pi_i \leq 1$ Constraints on response



We need a function mapping π_i onto a line.

$$- \text{odds} = \frac{\pi_i}{1-\pi_i} \in (0, \infty)$$

$$- \log \text{ odds} = \ln \frac{\pi_i}{1-\pi_i} \in (-\infty, \infty)$$

$$\text{logit}(\pi_i) = \ln \frac{\pi_i}{1-\pi_i} = X_i^T \beta$$

link function
maps π_i onto $X_i^T \beta$

linear predictor

$$\pi_i = \frac{e^{X_i^T \beta}}{1 + e^{X_i^T \beta}} = \frac{1}{1 + e^{-X_i^T \beta}}$$

logistic / sigmoid function